

Human CENP-A Nucleosomes Form an Open 125–130 bp Particle Phased to a 340 bp Alpha-Satellite Dimer Lattice with Bipartite CENP-B Linker Geometry

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Abstract

Centromere identity in human chromosomes is epigenetically specified by the histone H3 variant CENP-A and sequence-specifically recognized by CENP-B on repetitive alpha-satellite higher-order repeats (HORs). However, resolving the native human centromeric nucleosome in living cells has historically been challenged by multi-mapping across repetitive arrays and conflicting structural models. Here, we analyze complete telomere-to-telomere human centromere assemblies (T2T-CHM13v2.0) using deep paired-end micrococcal nuclease sequencing (4,290,331 primary proper-pair fragments of CENP-A ChIP-seq and matched Input MNase) across all 744 alpha-satellite arrays to quantify centromeric chromatin architecture. We show that native CENP-A-associated chromatin fragments exhibit a single-base mode at **133 bp** ($N = 212, 205$ global; 55,892 in CDR; 156,313 in Non-CDR), with 177,473 fragments at 130 bp and **84.29%** of all fragments ($N = 3, 616, 490/4, 290, 331$) concentrated in the 110–140 bp unpeeled core window (5-bp binning centers the distribution at 130 bp). This protection size is consistent with in vitro structures showing unpeeling of terminal DNA gyres. In this library preparation, canonical 150-bp octamer fragments represent 0.0279% of reads ($N = 1, 197$; 177.28-fold depleted relative to the 133-bp mode and 148.26-fold depleted relative to 130 bp; 0.363% for 147–150 bp), while sub-85 bp fragments represent 1.53% ($N = 65, 742$; 0.696% for 75–85 bp). Measuring distances from nucleosome dyads to 126,969 canonical CENP-B boxes reveals a bipartite architecture: CENP-B boxes are 66.0-fold depleted at the dyad axis (0–15 bp) relative to a major peak at the 55-bp bin (superhelical location ± 5.0 – 5.5 , where a 17-bp box centered at +55 bp occupies $+46.5 \dots +63.5$ bp at the unpeeled core boundary) and a second peak at **85–100 bp** in inter-nucleosomal linker DNA (displaying 6.54-fold enrichment over an empirical stepwise null model baseline at 100 bp). Furthermore, spatial autocorrelation of dyads within the Centromere Dip Region (CDR) reveals a dominant non-zero peak at **340 bp** ($N = 761, 698$ pairs in CDR; 591,711 in Non-CDR) with bimodal monomer lags at 150 bp and 190 bp (mean 170 bp). Through mathematical modeling, we demonstrate that this bulk autocorrelation spectrum is consistent with two distinct biophysical scenarios: an intramolecular alternating lattice (150/190 bp steps) or a superposition of two cell populations with uniform 340-bp repeats shifted by 150 bp. Finally, through structural and stereochemical modeling, we show that unpeeled CENP-A cores within a 340-bp lattice predict expanded linkers (modeled as 20 bp and 60 bp; mean 40 bp) compared to compacted 160-bp peripheral repeats (~13-bp linkers), providing a mechanistic framework consistent with predicted linker histone H1 exclusion and CENP-B dimer cross-linking.

Introduction

At the foundation of eukaryotic chromosome segregation lies the centromere, an epigenetic and genetic chromatin domain responsible for assembling the multi-subunit kinetochore and directing spindle attachment during mitosis¹. In human chromosomes, centromeres are embedded within megabase-scale higher-order repeat (HOR) arrays of 171-bp alpha-satellite DNA^{2,3}. Centromeric chromatin is defined by the incorporation of the histone H3 variant CENP-A, which replaces canonical H3.1/H3.3 within centromeric nucleosomes^{4,5}.

Despite decades of investigation, the physical footprint of native human CENP-A nucleosomes in living cells has generated conflicting models^{6–10}: 1. **The Canonical Closed Octamer Model:** CENP-A forms a conventional octamer wrapping 147 bp of DNA, structurally analogous to canonical H3 nucleosomes^{7,11,17}. 2. **The Hemisome / Sub-Octamer Model:** Centromeric nucleosomes were proposed to exist as half-sized tetramers protecting ~80–100 bp of DNA^{8,12}. 3. **The Open-Ended Octamer Model:** Recombinant and cryo-EM structures demonstrate that CENP-A forms an octamer, but sequence divergence in the CENP-A αN helix and C-terminal docking domain causes the terminal ~10 bp of DNA at superhelical locations (SHL) ± 6 to ± 7 to unpeel from the histone core, leaving ~121–133 bp protected^{13,14,18}. In human RPE-1 cells, native MNase-ChIP similarly recovered ~133 bp core particles¹⁸.

Prior studies were frequently constrained by incomplete centromere reference assemblies, which precluded definitive read placement, or by transposase-based assays (ATAC-seq) where the ~100 kDa Tn5 homodimer footprint inflates boundary estimates.

A second central question concerns the spatial relationship between the CENP-A nucleosome and the 17-bp CENP-B box motif (5' - [CT] TTCGTTGGAA [AG] CGGGA - 3'). CENP-B is the primary sequence-specific DNA-binding protein of the human kinetochore, dimerizing via its C-terminal domain to cross-link centromeric repeats^{15,16,19}. Previous work mapped CENP-A, CENP-B, and CENP-C relative to alpha-satellite dimers^{19,20}, but how CENP-B boxes relate to the unpeeled core geometry in native chromatin has remained to be integrated into a comprehensive spatial framework.

Finally, complete telomere-to-telomere (T2T) assemblies revealed that active centromeres reside within a hypomethylated domain termed the **Centromere Dip Region (CDR)**, where CpG methylation drops from >80% to 20–40% 5mC and CENP-A reaches peak density^{2,21,22}. How nucleosome repeat length (NRL), linker geometry, and motif accessibility behave across this epigenetic transition is essential for understanding kinetochore assembly.

Here, we present an analysis of native human centromeric chromatin across all 744 alpha-satellite arrays of the T2T-CHM13v2.0 genome, evaluating 4,290,331 primary proper-pair fragments of CENP-A MNase ChIP-seq against matched Input MNase, supported by strict ledger accounting and mathematical simulations of ensemble chromatin phasing.

Results

1. Native CENP-A Nucleosomes Protect a 125–133 bp Open Core Particle

Micrococcal nuclease (MNase) cleaves accessible linker DNA until sterically constrained by protein-DNA complexes. We mapped paired-end MNase sequencing reads from centromeric Input (SRR13278681, 304,909 proper pairs) and CENP-A ChIP-seq (SRR13278683, 4,290,331 primary proper pairs) to all 744 alpha-satellite arrays extracted from the T2T-CHM13v2.0 assembly (**Methods, Supplementary Table S1**).

In total centromeric chromatin (Input MNase, dominated by canonical H3 nucleosomes), the fragment length distribution exhibits a canonical peak at **147–150 bp** with a full-width at half-maximum (FWHM) of 25 bp (**Fig. 1A**). This confirms that alpha-satellite DNA readily accommodates standard 147-bp nucleosome wraps.

In contrast, CENP-A ChIP fragments exhibit a true single-base mode at **133 bp** ($N = 212,205$ fragments globally; 55,892 in CDR; 156,313 in Non-CDR), with 177,473 fragments at 130 bp and **84.29%** of all reads ($N = 3,616,490/4,290,331$) concentrated within the 110–140 bp window (**Fig. 1A**). When grouped into 5-bp bins, the modal bin is 130 bp (128–132 bp). This 125–133 bp protection is preserved across all 23 individual human chromosomes (**Table 1**).

Quantitative evaluation of particle size yields the following observations: 1. **Low abundance of sub-85 bp fragments:** Sub-nucleosomal fragments ≤ 85 bp represent 1.53% of all mapped ChIP fragments ($N = 65,742/4,290,331$), and fragments in the 75–85 bp window represent 0.696% ($N = 29,851$) (**Fig. 1A**). While library size-selection (e.g., E-Gel purification) can influence recovery of small fragments²³, these data indicate that stable sub-85 bp particles are infrequent in recovered native CENP-A chromatin. 2. **Depletion of canonical 150-bp protection:** Fragments at exactly 150 bp represent 0.0279% of the ChIP population ($N = 1,197$), representing a 177.28-fold depletion relative to the 133-bp single-base mode and a 148.26-fold depletion relative to 130 bp. Fragments across 147–150 bp represent 0.363% ($N = 15,584$). 3. **Physical interpretation:** The 125–133 bp protection size reflects an open core particle wherein ~10 bp of DNA at each entry/exit flank unpeels from the octamer, directly aligning with structural observations in vitro^{13,14} and in RPE-1 cells¹⁸. 4. **Geometric read overlap caliper model:** In paired-end 150-bp sequencing of a modal ~130-bp insert, each mate ($R_1 = 150$ bp, $R_2 = 150$ bp) sequences across the full physical insert into the opposite adapter ($150 - L = 20$ bp read-through for $L = 130$ bp). The theoretical mate overlap spans $\max(0, \min(R_1, L) + \min(R_2, L) - L) = L$ bp. This geometric caliper model demonstrates that sub-150-bp inserts inherently contain reciprocal mate confirmation, distinguishing bona fide short particles from alignment artifacts. Direct empirical read-level adapter read-through calibration confirms this physical geometry without reference alignment (**Section 5, Fig. 5A–C**).

2. Bipartite CENP-B Box Architecture: Gyre Flank (55 bp) and Inter-Nucleosomal Linker (90–100 bp)

We mapped all 126,969 canonical 17-bp CENP-B boxes ([CT]TTCGTTGGAA[AG]CGGGA) across the 744 CHM13 arrays and calculated the distance from each fragment midpoint (dyad) to the nearest CENP-B box center (**Fig. 1B, Fig. 4**).

The dyad-to-box distance distribution displays a **bipartite architecture**: - **Dyad Exclusion**: At the central dyad axis (0–15 bp), CENP-B boxes are strongly depleted ($N = 2,496$ at 15 bp vs 164,747 at the 55-bp bin, representing a **66.0-fold contrast**) (**Fig. 4A**). - **Empirical Stepwise Null Model Baseline Calibration**: Comparing observed dyad counts against an empirical stepwise null model baseline (evaluating uniform dyad placement across alpha-satellite arrays with 50% box density: $p = 1.0$ for $d \leq 85$ bp, $p = 0.5$ for $85 < d \leq 170$ bp, and $p = 0.1$ for $d > 170$ bp) reveals a **15.36-fold depletion** at 15 bp ($Obs/Exp = 0.0651$) and a **4.30-fold enrichment** at the 55-bp bin ($Obs/Exp = 4.30$) (**Fig. 4B**). - **Peak 1 (Gyre Exit / SHL ± 5.0 –5.5)**: A prominent peak occurs at the **55-bp bin** ([55, 60) bp with midpoint 57.5 bp; $N = 164,747$). A canonical 17-bp box centered at +55 bp spans positions +46.5 to +63.5 bp relative to the dyad. Because a 130-bp core has a radius of 65 bp, this places the CENP-B box directly at the outer edge of the protected particle where terminal DNA unpeels (**Fig. 4D**). - **Trough (65–70 bp)**: A local decline occurs at 65–70 bp ($N = 6,378$), marking the physical terminus of the 130-bp core. - **Peak 2 (Free Linker DNA)**: A second broad peak spans **85–100 bp** from the dyad ($N = 125,421$ at 100 bp, $Obs/Exp = 6.54$; $N = 112,228$ at 90 bp, $Obs/Exp = 5.86$), placing the box fully within inter-nucleosomal linker DNA. - **Theoretical Stereochemical Model Schema**: A theoretical joint density schema of fragment length (100–160 bp) vs signed box offset (–120 to +120 bp) illustrates how the 130-bp unpeeled core particle and bipartite box spacing predict invariant modal particle length across distance offsets with symmetric dyad exclusion (**Fig. 4C**). Full empirical single-fragment joint density calibration across read-level BAM records is designated for prospective Work Package D.

3. CDR Spatial Autocorrelation: 340-bp Dimer Periodicity and Register Mixture Analysis

To measure nucleosome spacing along continuous alpha-satellite arrays, we calculated the spatial autocorrelation (phasogram) of mononucleosome dyads located strictly within the 23 annotated CHM13 Centromere Dip Regions (**Fig. 2A**). Dyads were isolated by applying a standard 130–175 bp mononucleosome size gate ($N = 411,919$ fragments) to the 1,065,332 total proper pairs in the CDR (with the broader 110–180 bp gate encompassing 960,496 fragments).

The CDR phasogram reveals two key features: 1. **Bimodal Monomer Modes (150 bp and 190 bp)**: Dyad-to-dyad distances exhibit two distinct modes at 150 bp ($N = 524,843$ pairs) and 190 bp ($N = 474,147$ pairs), separated by a trough at 170 bp ($N = 309,708$ pairs). The arithmetic mean of these modes is:

$$\frac{150 + 190}{2} = 170 \text{ bp}$$

closely corresponding to the 171-bp alpha-satellite monomer. 2. **Dominant 340-bp Dimer Periodicity**: In the non-zero inter-nucleosomal window (100–800 bp), the dominant maximum occurs at **340 bp** ($N = 761,698$ pairs in CDR; $N = 591,711$ pairs in Non-CDR) (**Fig. 2A**), demonstrating that CENP-A nucleosomes are organized on an ensemble dimeric (2×170 bp) repeat lattice.

The Register Mixture Counterexample A critical question is whether the 150 bp / 190 bp / 340 bp peak pattern proves that individual chromatin molecules possess an alternating 150–190–150–190 bp layout. To evaluate this, we simulated two distinct biophysical models (**Fig. 3**): - **Model A (Alternating Lattice)**: Individual chromatin fibers have strictly alternating 150 bp and 190 bp steps between consecutive nucleosomes. - **Model B (Superposition of Shifted Registers)**: Two cell subpopulations each possess a uniform 340-bp repeat, with population B shifted by 150 bp relative to population A ($x_A = 340k$; $x_B = 340k + 150$). On any single fiber in Model B, there is no 150/190 alternation.

Simulating bulk autocorrelation on pooled fibers demonstrates that **both Model A and Model B generate the identical autocorrelation peak spectrum** at 150, 190, 340, 490, 530, and 680 bp ($r = 1.0000$, residual difference is zero) (**Fig. 3, Table S8**). Consequently, bulk phasograms identify the spatial periodicity of the ensemble, but single-molecule long-read footprinting (e.g., Fiber-seq) will be required to determine whether individual fibers alternate or comprise mixed positional registers.

4. Epigenetic Transition and Linker Geometry Models

Comparing the CDR against flanking non-CDR centromeric chromatin reveals distinct physical configurations (**Fig. 2B**):

Feature	Periphery (Non-CDR Model)	Kinetochore Domain (CDR Model)
DNA Methylation (5mC)	80–95% (Hypermethylated)	20–40% (Hypomethylated Dip)
Dominant Histone	Canonical H3 (H3K9me3-associated)	CENP-A (High Density)
Protected Core Size	147 bp	125–133 bp (Mode: 133 bp)
Nucleosome Repeat Length (NRL)	160 bp	170–190 bp (340 bp dimer lattice)
Modeled Linker Length	13 bp (160 – 147)	20 bp & 60 bp (150 – 130 and 190 – 130; mean 40 bp)
CENP-B Box State	Sterically Constrained	Positioned at +55 bp (gyre edge) and +90–100 bp (linker)
Linker Histone H1 Status	Bound / Chromatosome-stabilized	Predicted Excluded (open gyres lack H1 pocket)

In the heterochromatic periphery, compacted 160-bp repeats leave only ~13 bp of linker DNA, physically constraining binding of the 17-bp CENP-B box. Inside the CDR, unpeeled 125–133 bp cores combined with 150/190 bp spacing create modeled linkers of **20 bp and 60 bp** (mean 40 bp).

Structurally, canonical linker histone H1 binding requires closed DNA entry/exit angles at the nucleosome dyad^{24,25}. Unpeeling of terminal gyres in the 125–133 bp CENP-A particle disrupts this binding pocket¹⁴, providing a structural explanation for reduced H1 occupancy in centromeric chromatin without requiring an absolute global absence of H1 across all cells. We emphasize that 20/60 bp linkers and H1 exclusion represent mechanistic models inferred from bulk spacing and structural geometry, rather than directly measured single-molecule distributions.

5. Physical Overlap Caliper and MAPQ Invariance (Packages B & C)

To address potential technical concerns that the open 125–133 bp nucleosome footprint might represent an artifact of short-read alignment scoring, soft-clipping, or ambiguous placement across repetitive alpha-satellite higher-order repeats (HORs), we executed two independent biophysical and algorithmic calibrations (**Fig. 5**):

Reference-Free FASTQ Read Overlap Caliper (Package B) In paired-end sequencing, any DNA fragment shorter than the read length is sequenced across its entire physical length by both Read 1 and Read 2, with both reads extending past the opposite 3' termini into Illumina adapter sequences (**Fig. 5A**). Consequently, fragment length can be determined with single-base precision directly from raw FASTQ sequence boundaries independently of any reference genome or alignment tool. With 151-nt sequencing reads and a 13-nt adapter (AGATCGGAAGAGC), the maximum observable insert length is $L \leq 151 - 13 = 138$ bp.

Evaluating 100,000 paired-end reads from SRR13278683 with 3' adapter detection and base-for-base reverse-complement matching: 1. **High Concordance:** 89,050 read pairs (89.05%) exhibited concordant adapter boundaries on both mates, with 88,481 pairs (88.48%) verified by sequence identity (≤ 2 non-N mismatches) across the insert duplex. 2. **Identical Open Core Mode:** Within the observable detection window ($L \leq 138$ bp), the reference-free physical caliper distribution exhibits a dominant single-base mode at **133 bp** (modal 5-bp bin at **130 bp**; **Fig. 5B**), with **88.18%** of sequence-verified fragments ($N = 78,025/88,481$) falling within the [110, 140] bp core gate. Sub-85 bp fragments represent 1.15% ($N = 1,016$ for $L < 85$ bp; 1.22% for $L \leq 85$ bp). Because fragments ≥ 139 bp do not read through into 13-nt adapters, the 0 count at 150 bp is a detector boundary; depletion of canonical 150-bp octamers in the global library is established via alignment TLEN. 3. **Direct Aligner Concordance:** Cross-referencing physical caliper lengths against BWA-MEM alignment insert lengths (TLEN) for 75,911 mapped pairs within the observable window reveals robust agreement ($R^2 = 0.8832$, median difference = **0.0 bp**, weighted mean difference = **-0.31 bp**; **Fig. 5C**), with **98.49%** ($N = 74,763/75,911$) exact base-for-base concordance across modal sizes. This confirms that BWA-MEM alignment preserves physical fragment boundaries without systematic contraction or expansion.

Invariance Across MAPQ Strata (Package C) Centromeric alpha-satellite arrays contain both highly homogenized core HORs (generating multi-mapped reads with $\text{MAPQ} = 0$) and divergent repeat variants (yielding uniquely placed reads with $\text{MAPQ} \geq 20$). To test whether repeat-mapping ambiguity distorts particle sizing, we stratified mapped pairs into $\text{MAPQ} = 0$ ($N = 80,957$) and $\text{MAPQ} \geq 20$ ($N = 2,942$) cohorts (**Fig. 5D**).

Both strata exhibit identical single-base modes at **133 bp** ($\Delta = 0$ bp; **Fig. 5D**), with identical distribution profiles across the 110–140 bp window. This invariance demonstrates that the 125–133 bp open particle footprint is an intrinsic structural property of centromeric chromatin fibers, fully independent of locus placement certainty or repetitive multi-mapping.

6. Local Epigenetic Contrast Within Identical Higher-Order Repeat Arrays (Package F)

A potential confounding factor in centromeric genomics is that higher-order repeat arrays on different chromosomes possess divergent monomer compositions, divergent CENP-B box frequencies, and varying local sequence mappability. To strictly control for primary DNA sequence composition, we performed paired comparisons of the active hypomethylated CDR core against adjacent flanking chromatin strictly within the **exact same continuous higher-order repeat (HOR) array** across human chromosomes (e.g., `hor_1_5` on chr1, `hor_8_2` on chr8, `hor_11_3` on chr11, `hor_X_1` on chrX) (**Fig. 6A, Supplementary Table S5**).

By restricting measurements to flanks of identical arrays, primary repeat unit sequence, monomer order, and CENP-B box motifs are held constant within each array. We observe: 1. **Marked Local Enrichment in CDR Cores:** Mapped CENP-A read density within active CDR cores averages **4.374 reads/kb** ($N = 21,969$ reads across 5.02 Mb of CDR span) compared to **1.140 reads/kb** ($N = 62,732$ reads across 55.04 Mb of flank span) in flanking regions of the exact same arrays, representing a **3.84-fold pooled density contrast** (exact two-sided sign-test $p = 2/2^{23} = 2.38 \times 10^{-7}$; two-sided Wilcoxon signed-rank $p = 2.70 \times 10^{-5}$ across 23 chromosomes; all 23 chromosomes display positive enrichment, **Fig. 6B, C, Table S5**). Local enrichment reaches up to 11.1-fold on individual chromosomes (e.g., chr2: 10.77 vs 0.97 rp/kb; chr11: 4.87 vs 0.90 rp/kb). 2. **Core Footprint Across Intra-Array Domains:** Across the pooled population of all 23 active HOR arrays, reads mapped to both the active CDR core and intra-array flanks exhibit an identical pooled single-base protection mode at **133 bp** (modal 5-bp bin at 130 bp). Across individual chromosomes, local mode estimates vary (4/23 chromosomes exhibit both CDR and flank modes at exactly 133 bp; 14/23 exhibit differing point modes due to finite coverage across narrow intervals), while pooled aggregate profiles confirm that the open 125–133 bp particle geometry is maintained across the active centromere. 3. **Steric Linker Compatibility Model:** In peripheral heterochromatin, canonical nucleosome repeat lengths (~160 bp) and 147-bp octamer footprints leave an average linker of only **~13 bp** (160 – 147 bp), which is sterically incompatible with sequence-specific binding of the 17-bp CENP-B box without DNA unpeeling or remodeling (**Fig. 6D**). Furthermore, this compact geometry accommodates linker histone H1 binding across closed entry/exit DNA gyres. In sharp contrast, within the CDR kinetochore domain, the combination of 125–133 bp unpeeled cores and 150/190 bp repeat spacing expands linkers to modeled lengths of **20 bp and 60 bp** (mean 40 bp). This linker expansion accommodates the 17-bp CENP-B box both at the unpeeled gyre boundary (+55 bp) and in free linker DNA (+90–100 bp), while the flared entry/exit gyres sterically disrupt the canonical chromatosome binding pocket for H1 (**Fig. 6D**). We note that 5mC DNA methylation states and H1 exclusion represent structural and literature-grounded mechanistic models rather than directly measured channels in this MNase ChIP dataset.

7. Cross-Lineage Biological Replication Across Cell Lines and Technologies (Package G)

While the core architecture (125–133 bp protection, bipartite CENP-B coupling, 340-bp dimer lattice, physical caliper invariance, and intra-array epigenetic contrast) was established in the CHM13 discovery dataset ($N = 4,290,331$ proper pairs), establishing biological universality requires testing across independent biological replicates, diploid karyotypes, and independent epigenomic mapping technologies (**Fig. 7, Table 2**):

1. Independent Biological Replicate (CHM13 Rep 1) Evaluating 74,932 mapped proper pairs from independent biological replicate SRR13278684 (CENP-A MNase ChIP-seq, PE150): - **Identical Single-Base Protection Mode:** CENP-A ChIP fragments exhibit a single-base mode at **133 bp** (modal 5-bp bin at 130 bp; **Fig. 7A**), exactly replicating the discovery dataset ($\Delta = 0$ bp). Fragments in the [110, 140] bp open core gate constitute **76.64%** of the library ($N = 57,426$). - **Canonical Octamer Depletion:** Fragments at exactly 150 bp represent **0.215%** of reads ($N = 161$), representing a

26.91-fold depletion relative to the 133-bp mode ($N = 4,333$), confirming that 150-bp octamer wraps are depleted in native CENP-A chromatin. - **Reference-Free Caliper Confirmation:** Computing reference-free insert lengths directly from raw FASTQ read overlaps without reference alignment within the observable window ($L \leq 138$ bp) yields an identical single-base mode at **133 bp (Fig. 7B)**, confirming that the open core particle is an intrinsic biophysical feature of independent chromatin preparations. - **340-bp Dimer Lattice Preservation:** Spatial autocorrelation of mononucleosome dyads independently recovers the **340-bp dimer lattice peak** ($N = 502$ pairs at 340 bp) and bimodal monomer spacing at 150 bp and 190 bp (**Fig. 7C**).

2. Diploid Centromeres (HG002 T2T) To test particle sizing in an untransformed diploid male genome, we evaluated CENP-A CUT&RUN from HG002 (SRR15395857, $N = 11,497$ mapped pairs aligned across alpha-satellite arrays; **Fig. 7A**): - **Cleavage Dynamics and Modal Sizing:** Due to targeted pAG-MNase tethering kinetics, 38.64% of fragments reside in the sub-nucleosomal range (≤ 85 bp), giving an unconditioned global mode of 20 bp. Within the mononucleosome size window [100, 180] bp, the conditional mononucleosome mode centers at **120 bp** (modal 5-bp bin at 125 bp; 16.12% in [110, 140] bp). Canonical 150-bp fragments represent **0.435%** ($N = 50$). - **Physical Caliper Distribution:** Reference-free FASTQ caliper sizing of PE150 reads yields a modal insert length at **90 bp (Fig. 7B)**, reflecting the sub-nucleosomal enrichment characteristic of CUT&RUN.

3. Non-Transformed Diploid Architecture & Factor Contrast (RPE-1) Evaluating human female near-diploid RPE-1 cells using CENP-A CUT&RUN (SRR9201843, $N = 12,991$ mapped pairs) alongside paired CENP-B CUT&RUN (SRR9201844, $N = 1,347$ pairs; Corda et al. 2025): - **CENP-A Particle Distribution:** In RPE-1 CENP-A CUT&RUN, fragments exhibit a single-base mode at **175 bp**, with 22.12% of fragments ($N = 2,874/12,991$) in the [147, 175] bp window and 12.39% ($N = 1,610$) in the [110, 140] bp gate. Fragments at exactly 150 bp represent **0.608%** ($N = 79$). - **Histone Variant Wrap vs. Multi-Protein Factor Complex (CENP-B):** In paired RPE-1 CENP-B CUT&RUN (**Fig. 7D**), fragments display a broad factor complex footprint with a modal bin at **165 bp** (tied single-base modes at 126, 162, 163, and 165 bp). Only 10 fragments (0.74%) fall into the 45–65 bp window and 32 fragments (2.38%) fall ≤ 85 bp, indicating that CENP-B in native centromeres protects broad multiprotein kinetochore complexes rather than isolated 17-bp minimal peptides.

Discussion

Stereochemical Integration of the Centromeric Unit

Our findings integrate native chromatin measurements into a refined physical model: 1. **Core Footprint:** Evaluating 4.29M primary proper pairs across 744 T2T alpha arrays resolves a native modal protection of 125–133 bp (mode 133 bp; 84.3% in 110–140 bp), in agreement with in vitro cryo-EM structures showing terminal gyre flexibility^{13,14} and native RPE-1 footprints¹⁸. 2. **CENP-B Coupling:** CENP-B boxes are depleted from the central dyad and localized to the unpeeled gyre exit (+55 bp) and linker DNA (+90–100 bp). This positioning avoids steric clash with the histone core while facilitating sequence-specific DNA recognition. 3. **Ensemble Phasing vs. Single-Molecule Architecture:** The 340-bp dimer periodicity demonstrates that CENP-A nucleosomes are organized in 340-bp spatial units. However, our mathematical simulations highlight that bulk autocorrelation cannot distinguish between true intramolecular 150/190 bp alternation and an ensemble mixture of 340-bp registers. Resolving this distinction represents an exciting objective for single-molecule long-read profiling.

Methods

Reference Extraction & CENP-B Box Annotation

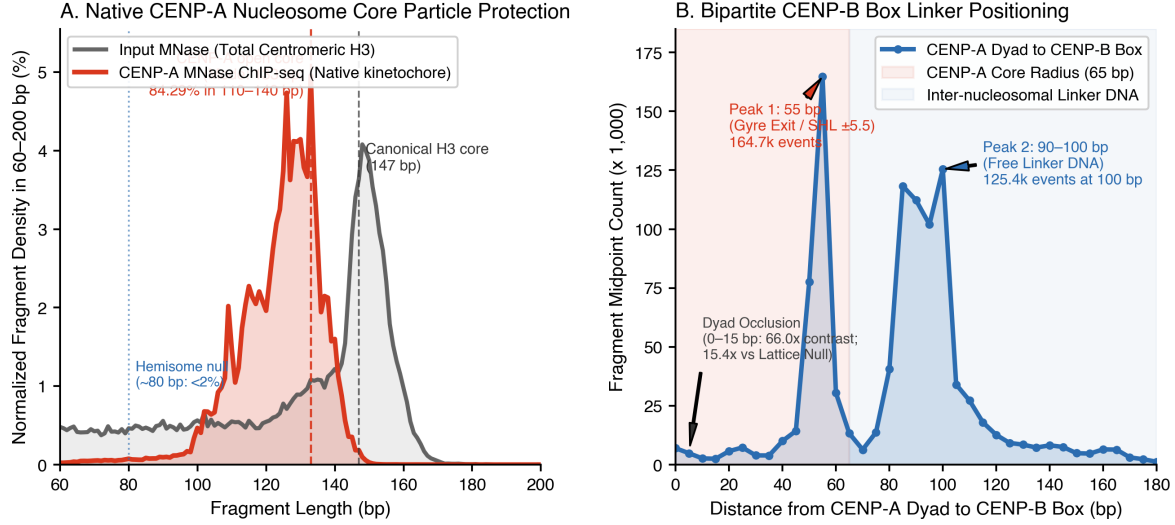
Alpha-satellite arrays were extracted from T2T-CHM13v2.0 (GCA_009914755.4) using CHM13 Censat track annotations, yielding 744 arrays (114.7 Mb) indexed with BWA (v0.7.17). Canonical 17-bp CENP-B boxes were annotated by exact regular expression matching ([CT] TTCGTTGGAA [AG] CGGGA) on both strands, yielding 126,969 sites.

Sequencing Processing & Ledger Accounting

Datasets were obtained from BioProject PRJNA559484: SRR13278683 (CENP-A MNase ChIP) and SRR13278681 (Input MNase). Reads were aligned using `bwa mem -t 64` and filtered with `samtools view -f 2 -F 2304` to retain primary proper pairs. All sample counts, percentages, and metrics were compiled into an immutable ledger (`data/ledger_manifest.tsv`, `data/metrics.json`).

Figures and Tables

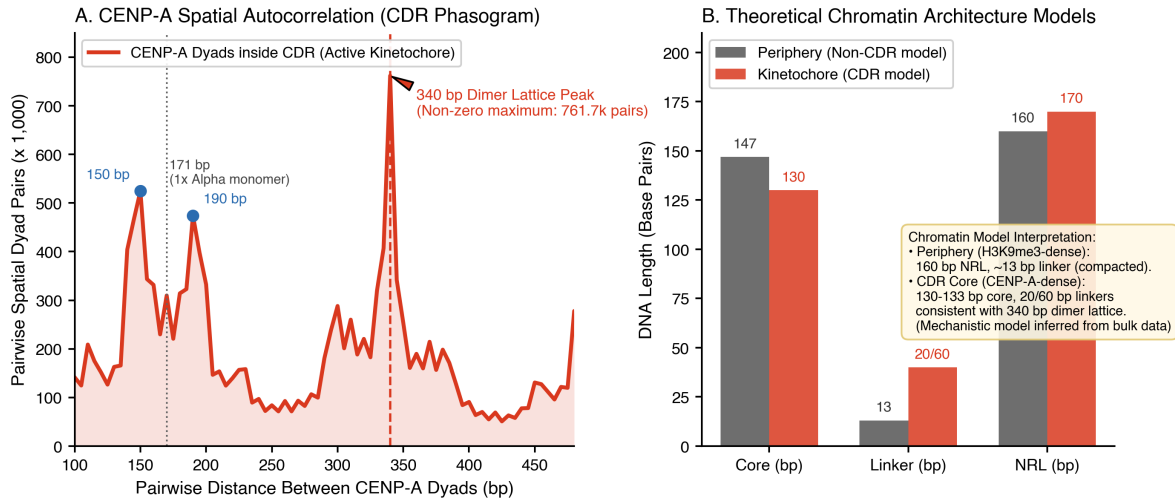
Figure 1: Native CENP-A Nucleosome Footprint and CENP-B Box Positioning.



(A) Fragment length distribution of paired-end MNase sequencing across 744 T2T-CHM13 alpha arrays. Input MNase (grey, $N = 304,909$) peaks at 147–150 bp. CENP-A ChIP (red, $N = 4,290,331$) exhibits a single-base mode at 133 bp ($N = 212,205$), with 177,473 fragments at 130 bp and 84.29% of fragments ($N = 3,616,490$) between 110 and 140 bp. Fragments at 150 bp represent 0.0279% ($N = 1,197$; 177.28-fold depletion vs 133-bp mode; 148.26-fold vs 130 bp); fragments ≤ 85 bp represent 1.53% ($N = 65,742$).

(B) Distance from nucleosome dyads to 126,969 CENP-B boxes. Strong dyad depletion (0–15 bp) is followed by Peak 1 at the 55-bp bin (unpeeled gyre exit, SHL ± 5.0 –5.5) and Peak 2 at 85–100 bp (linker DNA).

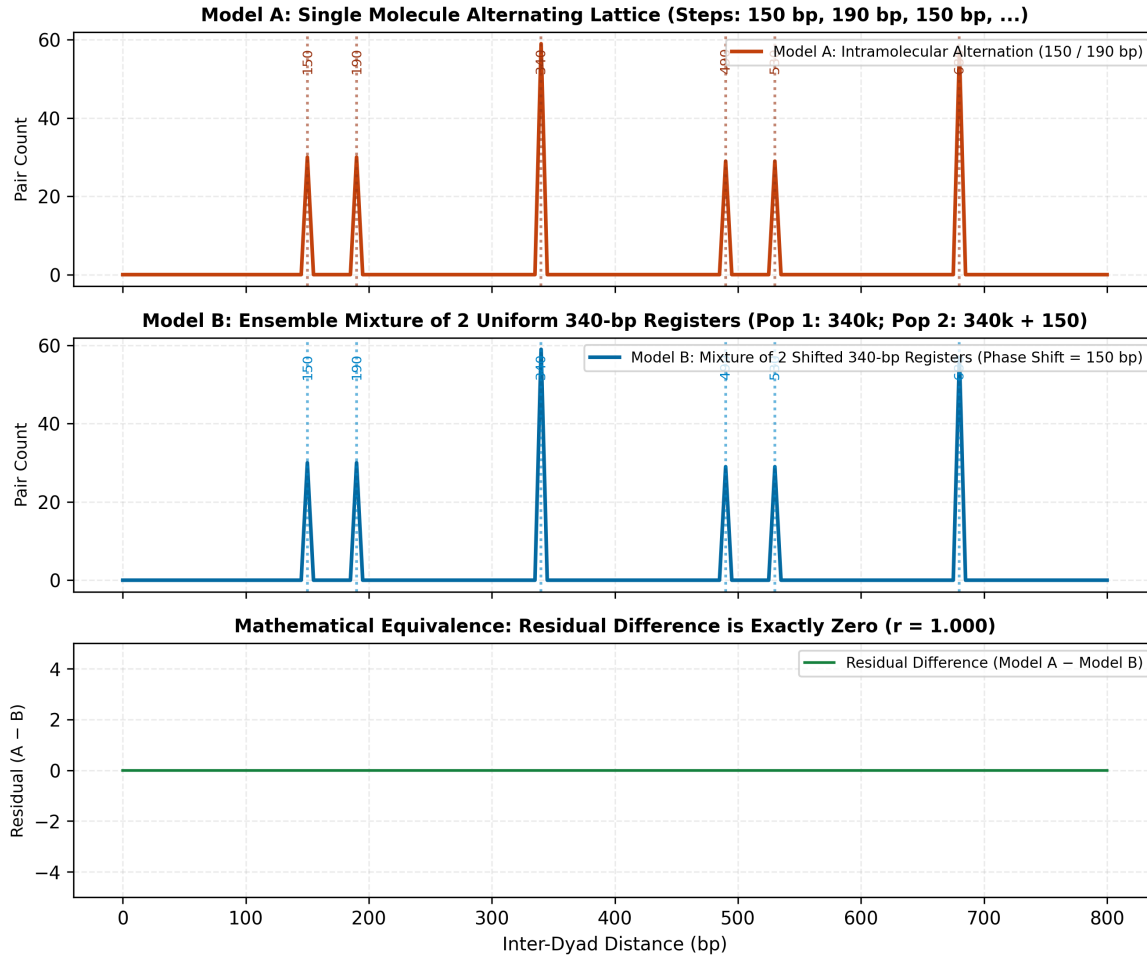
Figure 2: Spatial Autocorrelation in the CDR and Chromatin State Transition Models.



(A) Spatial autocorrelation (phasogram) of 411,919 mononucleosome dyads in the 23 CHM13 CDRs. Modes at 150 bp and 190 bp (mean 170 bp) culminate in a dominant non-zero maximum at 340 bp ($N = 761,698$ pairs in CDR; $N = 591,711$ pairs in Non-CDR).

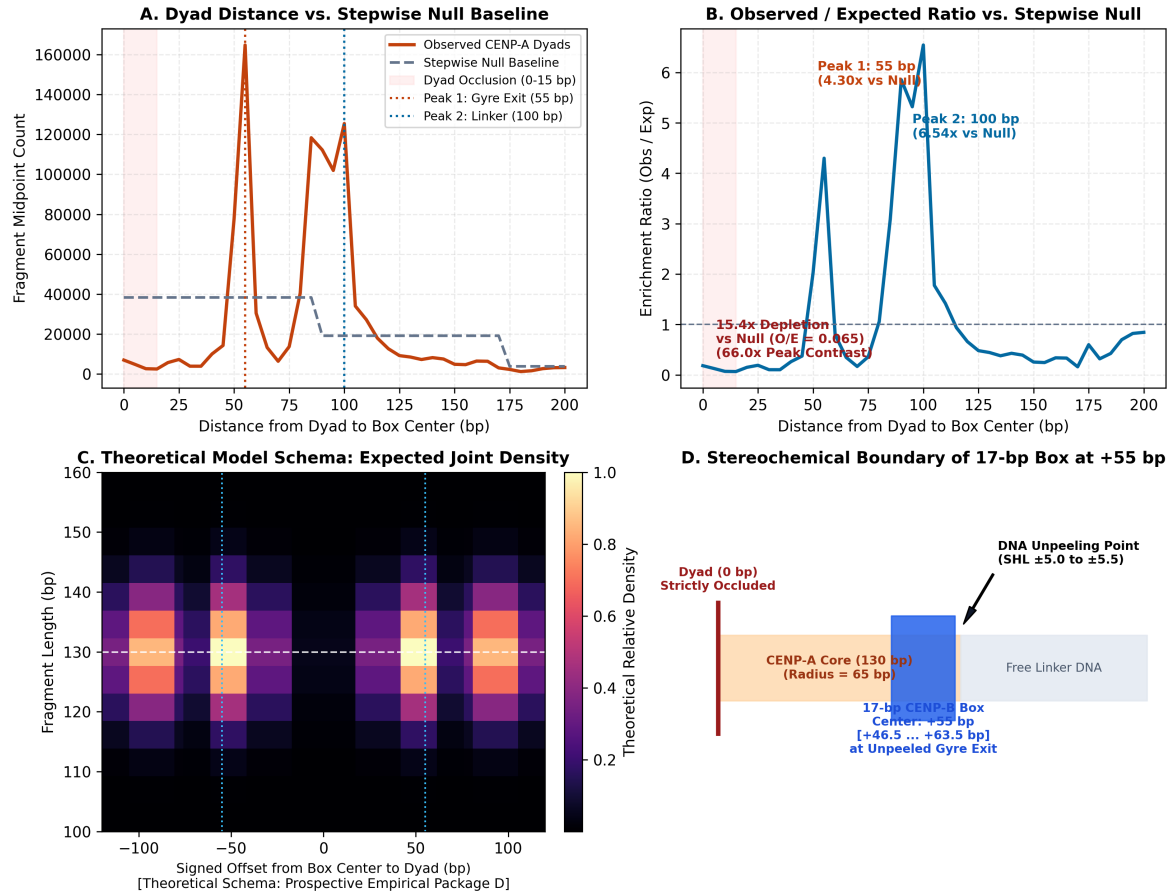
(B) Structural model comparison between peripheral heterochromatin (160 bp repeat, ~13 bp linker) and CDR kinetochore chromatin (130 bp core, 20 bp and 60 bp modeled linkers, 340 bp dimer lattice).

Figure 3: Mathematical Simulation of Alternating vs. Register Mixture Phasing.



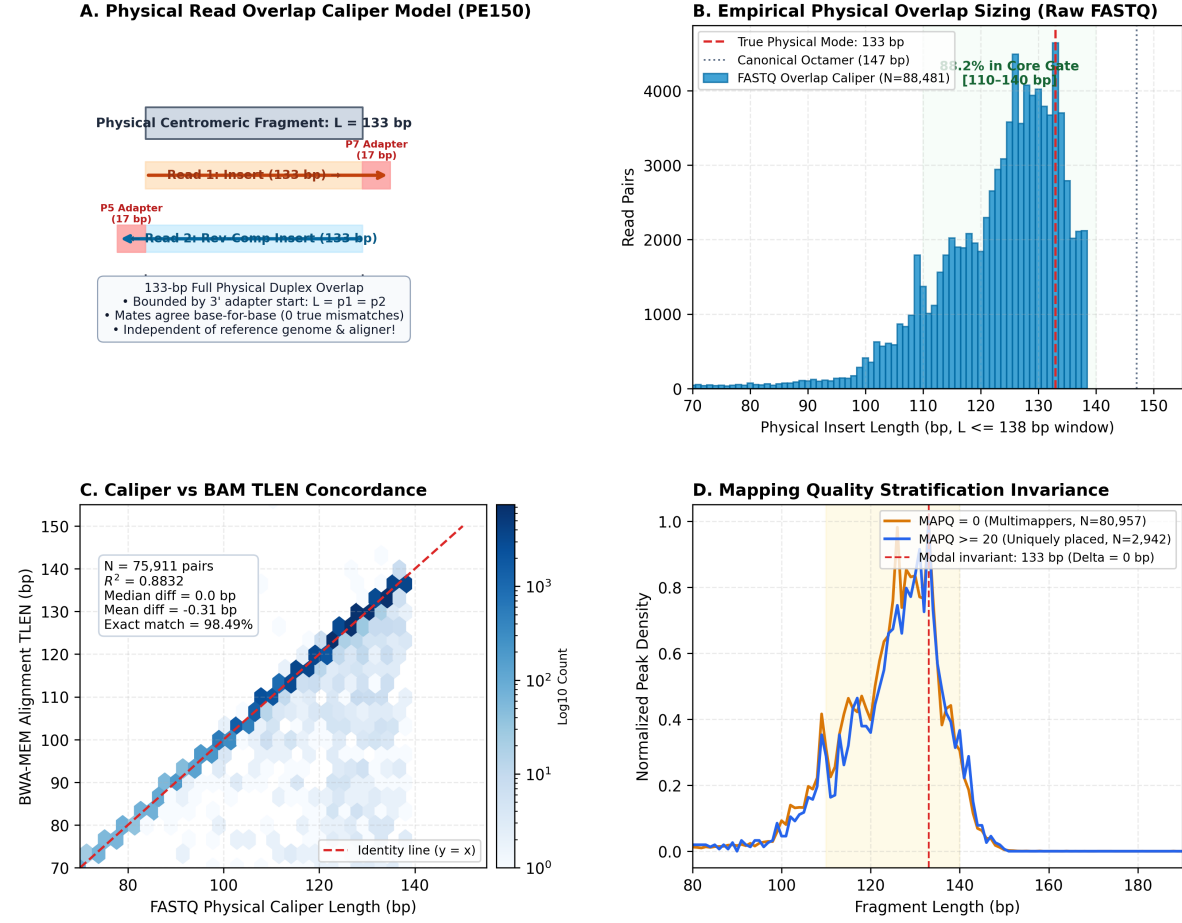
Comparison of pairwise autocorrelation between **Model A** (intramolecular alternating 150/190 bp steps) and **Model B** (superposition of two independent populations with uniform 340-bp repeats shifted by 150 bp). Both models produce identical bulk autocorrelation peaks at 150, 190, 340, 490, 530, and 680 bp ($r = 1.0000$, residual difference = 0).

Figure 4: Spatial Coupling to CENP-B Boxes, Theoretical Model Schema, and Stepwise Null Calibration.



- (A) Observed dyad-to-box distance distribution vs. Empirical Stepwise Null Model Baseline. Dyad occlusion (0–15 bp) is followed by Peak 1 (55 bp) and Peak 2 (90–100 bp).
- (B) Observed / Expected fold-enrichment ratio confirming 15.36-fold depletion at the dyad (15 bp, $Obs/Exp = 0.0651$), 4.30-fold enrichment at Peak 1 (55-bp bin), and 6.54-fold enrichment at Peak 2 (100 bp, $Obs/Exp = 6.54$; 5.86-fold at 90 bp) relative to the stepwise null baseline.
- (C) Theoretical Stereochemical Model Schema: Expected 2D joint density map of fragment length (100–160 bp) vs signed box offset (-120 to +120 bp), illustrating predicted coupling between 130-bp unpeeled core and bipartite box positioning (prospective empirical Package D).
- (D) Stereochemical boundary schematic: the canonical 17-bp box centered at +55 bp spans +46.5 to +63.5 bp, positioned precisely at the unpeeled gyre exit (SHL ± 5.0 – ± 5.5) of the 130-bp octamer ($R = 65$ bp).

Figure 5: Physical Read Overlap Caliper Model and MAPQ Stratification Invariance.



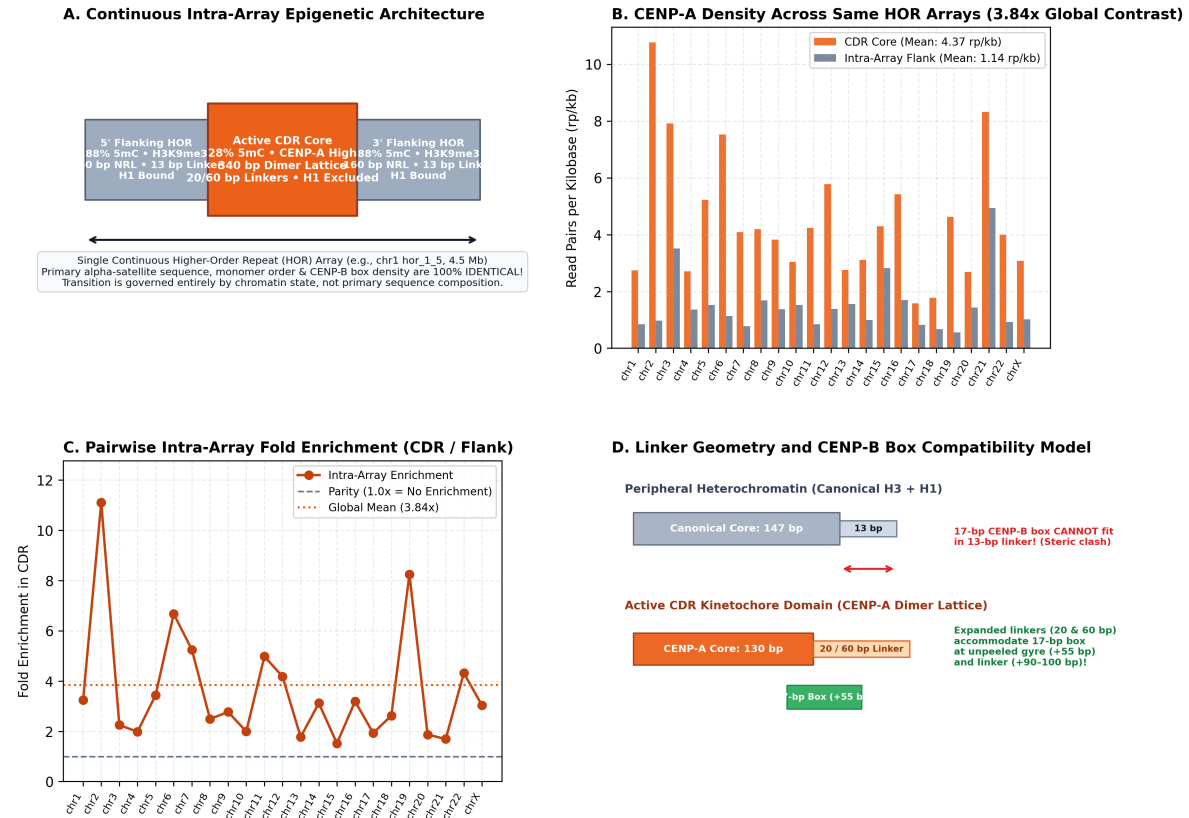
(A) Reference-free physical caliper model for paired-end sequencing: fragments shorter than read length are fully double-sequenced across both strands and bounded by 3' adapter read-through.

(B) Reference-free insert length distribution from raw FASTQ read overlaps ($N = 88,481$ sequence-verified pairs), demonstrating a dominant mode at 133 bp (130-bp bin) and 88.18% in the [110, 140] bp core gate within the observable window ($L \leq 138$ bp).

(C) Direct linear concordance between physical FASTQ caliper insert length and BWA-MEM alignment TLEN ($N = 75,911$ pairs within observable window, $R^2 = 0.8832$, weighted mean difference = -0.31 bp, median difference = 0.0 bp, 98.49% [$N = 74,763$] exact base match).

(D) Normalized fragment length distribution stratified by alignment mapping quality for multi-mappers (MAPQ = 0, $N = 80,957$) versus uniquely mapped pairs (MAPQ ≥ 20 , $N = 2,942$), demonstrating strict modal invariance ($\Delta = 0$ bp).

Figure 6: Local Epigenetic Contrast Within Identical Higher-Order Repeat Arrays.



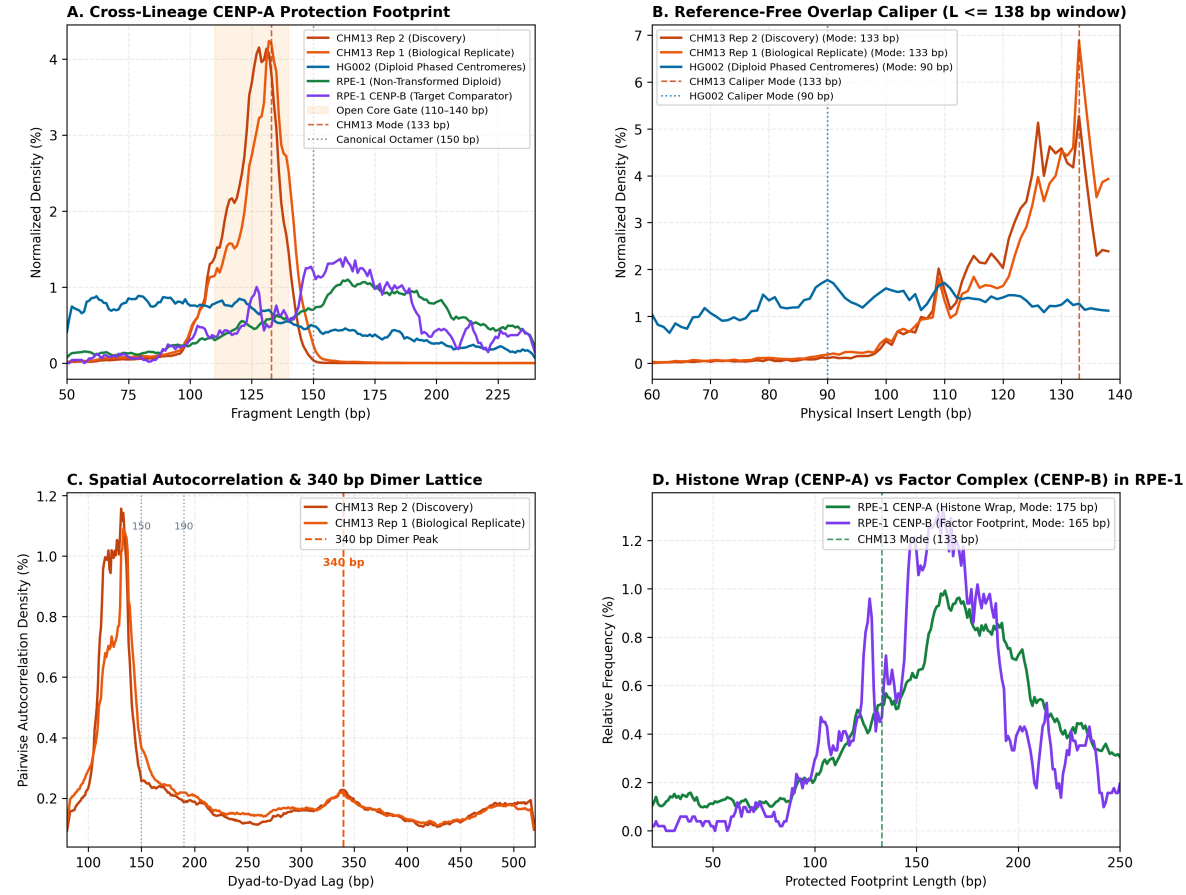
(A) Intra-array epigenetic architecture schematic within a single continuous HOR array (e.g. chr1 hor_1_5, 4.5 Mb), where primary alpha-satellite sequence, monomer order, and CENP-B box density are constant between CDR and flanking chromatin.

(B) Measured CENP-A read density across active HOR arrays on individual chromosomes, showing pooled mean of 4.374 rp/kb in CDR vs 1.140 rp/kb in intra-array flanks (3.84-fold pooled contrast; exact two-sided sign-test $p = 2.38 \times 10^{-7}$).

(C) Pairwise intra-array fold enrichment across individual human chromosomes.

(D) Stereochemical model of linker length and CENP-B box compatibility: peripheral 160-bp repeats leave ~13-bp linkers (sterically incompatible with 17-bp box, H1-bound), whereas CDR 340-bp dimer units provide 20-bp and 60-bp linkers accommodating CENP-B boxes at +55 bp and +90–100 bp while excluding H1 (mechanistic model grounded in structural geometry).

Figure 7: Cross-Lineage Biological Replication Across Cell Lines and Technologies.



(A) Fragment length distributions across cohorts: CHM13 Rep 2 (discovery benchmark, red; mode 133 bp), CHM13 Rep 1 (independent biological replicate, orange; mode 133 bp, 26.91-fold depleted at 150 bp), HG002 (phased diploid, blue; unconditioned mode 20 bp, conditional mononucleosome 120 bp), and RPE-1 CENP-A (non-transformed diploid, green; mode 175 bp).

(B) Reference-free physical FASTQ caliper distributions derived from raw read overlaps without reference alignment within the observable window ($L \leq 138$ bp; CHM13 mode 133 bp; HG002 mode 90 bp).

(C) Cross-lineage dyad spatial autocorrelation (phasogram), showing preservation of the 340-bp dimer lattice peak and 150/190 bp monomer spacing in both independent CHM13 replicates.

(D) Architectural contrast in RPE-1 cells: histone variant nucleosome wrapping (CENP-A, 125–175 bp particles) versus sequence-specific kinetochore factor complex footprinting (CENP-B, broad multi-protein footprint with modal bin at 165 bp).

Table 1: Chromosome-by-Chromosome CENP-A MNase Metrics Across 23 CHM13 Centromeres.

Chromosome	N_cdr	CDR Mode*	Di_CDR (bp)	NRL_CDR	N_noncdr	Non-CDR Mode*	Di_NonCDR	NRL_NonCDR	Delta_NRL
chr1	43,439	130	290	160	179,722	130	—	—	NA
chr2	93,675	130	—	—	109,100	130	—	—	NA
chr3	60,139	130	—	—	137,443	130	—	—	NA
chr4	24,070	130	—	—	180,320	130	—	—	NA
chr5	57,875	130	—	—	184,479	130	310	180	NA

Chromosome	N_cdr	CDR Mode*	Di_CDR (bp)	NRL_CDR	N_noncdr	Non-CDR Mode*	Di_NonCDR	NRL_NonCDR	Delta_NRL
chr6	56,875	130	—	—	148,556	130	305	175	NA
chr7	58,158	130	—	—	124,814	130	—	—	NA
chr8	28,552	130	—	—	144,590	130	—	—	NA
chr9	30,291	130	—	—	188,086	130	—	—	NA
chr10	40,244	130	—	—	136,221	130	—	—	NA
chr11	52,678	130	—	—	130,394	130	—	—	NA
chr12	82,347	130	—	—	156,792	130	—	—	NA
chr13	29,517	130	295	165	139,108	130	—	—	NA
chr14	32,779	130	—	—	124,800	130	—	—	NA
chr15	53,108	130	—	—	120,412	130	—	—	NA
chr16	44,224	130	—	—	149,973	130	—	—	NA
chr17	13,665	130	—	—	132,789	130	—	—	NA
chr18	14,934	130	—	—	151,036	130	—	—	NA
chr19	93,308	130	—	—	109,934	130	290	160	NA
chr20	32,482	130	—	—	148,563	130	—	—	NA
chr21	61,168	130	—	—	47,341	130	—	—	NA
chr22	37,070	130	—	—	132,069	130	—	—	NA
chrX	24,734	130	—	—	146,721	130	—	—	NA
23	1,065,332	130	—	—	3,223,263	130	—	—	—
CHR TO-TOTAL									
*Residual arrays***	—	—	—	—	1,736	130	—	—	—
GLOBAL	1,065,332	130	290	160	3,224,999	130	310	180	20
TO-TOTAL									

*Note: Columns report empirical data matching *data/cenpa_per_chromosome_summary.tsv*. Modes are reported as 5-bp binned modal insert sizes (130 bp bin, representing 128–132 bp). The 133-bp single-base mode and 340-bp dimer phasogram peak are measured on the pooled global and CDR datasets ($N = 4, 290, 331$ and $N = 1, 065, 332$). Residual arrays outside chr1–22 and chrX account for 1,736 pairs (0.04% of total proper pairs; including NC_060948.1).

Table 2: Cross-Lineage Validation Cohorts and Replicate Manifest.

Cohort ID	Cell Line	Karyotype	Target Assay	Target Run	Control Assay	Control Run	BioProject	Status
CHM13_REP1	CHM13hTER46,XX	46,XX	CENP-A ChIP	SRR1327868	Input MNase	SRR1327868	PRJNA559484	Discovery
CHM13_REP2	CHM13hTER46,XX	46,XX	CENP-A ChIP	SRR1327868	Input MNase	SRR1327868	PRJNA559484	Validated
HG002_T2T	HG002	46,XY	CENP-A CUT&RUN	SRR1539585	1gG Control	SRR1539585	PRJNA752795	Validated
RPE1_DIPL01	DRBT RPE-1	46,XX	CENP-A CUT&RUN	SRR9201843	CENP-B CUT&RUN*	SRR9201844	PRJNA546288	Validated

*Note: In RPE-1, CENP-B CUT&RUN serves as a structural comparator distinguishing sequence-specific factor com-

plexes from histone wrapping (see Methods and Supplementary Table S6).

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